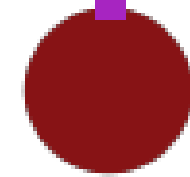


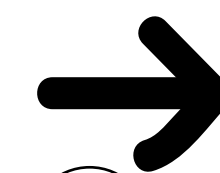
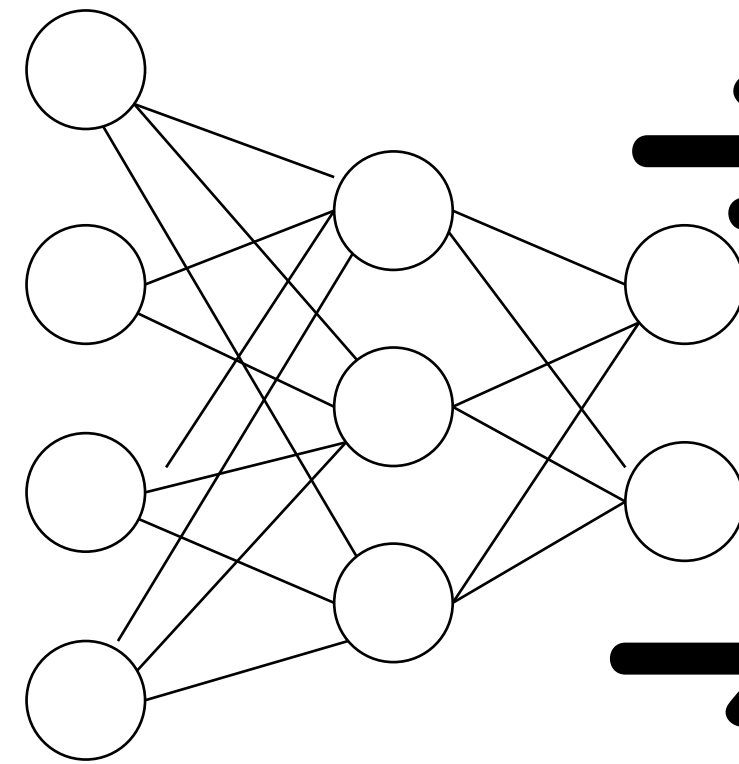
RL in Prey and Predators



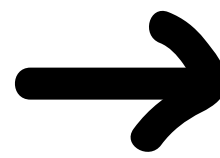
Sensory Inputs

- Touch Sensors
- Proximity Sensors
- Angle
- Velocity
- Angular Velocity
- Energy

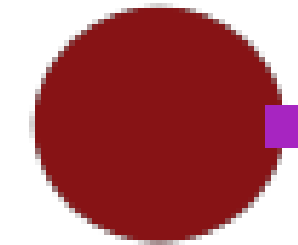
MLP



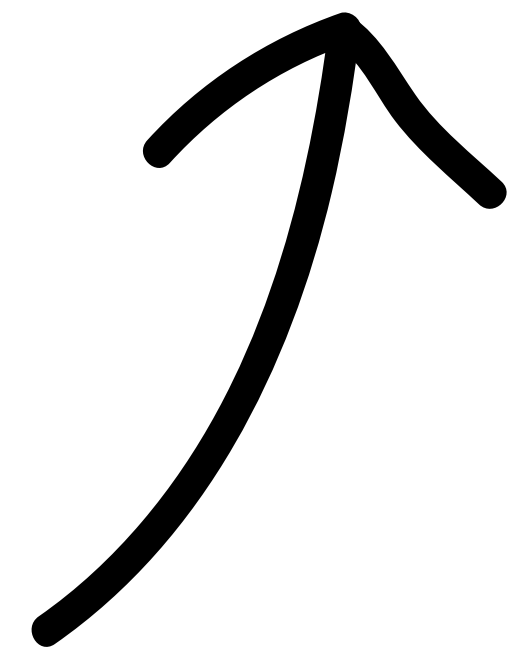
Left Motor Output



Right Motor Output



Learn to maximize
rewards via PPO



$$r_j^i = w_{\text{eat}}^i n_j^i + 0.01 w_{\text{act}}^i \frac{1}{F} \|f_j^i\|$$

$$+ 0.1 w_{\text{prey}}^i \max_k s_{\text{prey}}^{i,k} + 0.1 w_{\text{pred}}^i \max_k s_{\text{pred}}^{i,k}$$

Generate Reward by inherited weights